



ALLIANCE
UNIVERSITY

*Private University established in Karnataka State by Act No. 34 of year 2010
Recognised by the University Grants Commission (UGC), New Delhi*

DATABASE MANAGEMENT SYSTEM

CAB BOOKING MANAGEMENT SYSTEM

A relational database design for managing customers, drivers, cabs, bookings, trips, payments and ratings.

BACHELORS OF COMPUTER APPLICATION- III

NAME: PREKSHA V

REGISTER NUMBER: 2511021240063

SUBMITTING TO,

RENITA JOY R

DBMS ASSIGNMENT-2

1. Entities and Attributes

The application contains eight core entities: Customer, Driver, Cab, Booking, Trip, Payment, Rating and Driver_Location.

CUSTOMER

Attribute	Description	Key
Customer_ID	Unique customer identifier	PK
Name	Customer name	
Phone	Customer contact number	
Email	Customer email	
Address	Customer address	

DRIVER

Attribute	Description	Key
Driver_ID	Unique driver identifier	PK
Name	Driver name	
Phone	Driver contact number	
License_Number	Driving license number	
Status	Available / Busy / Offline	

CAB

Attribute	Description	Key
Cab_ID	Unique cab identifier	PK
Driver_ID	Driver assigned to the cab	FK
Vehicle_Number	Registration number	
Cab_Type	Sedan / SUV / Hatchback	
Model	Vehicle model	
Status	Available / Booked / Maintenance	

BOOKING

Attribute	Description	Key
Booking_ID	Unique booking identifier	PK
Customer_ID	Customer who made booking	FK
Driver_ID	Assigned driver	FK
Cab_ID	Assigned cab	FK
Booking_Date	Date and time of booking	

Attribute	Description	Key
Pickup_Location	Pickup point	
Drop_Location	Destination	
Booking_Status	Pending / Confirmed / Cancelled / Completed	

TRIP

Attribute	Description	Key
Trip_ID	Unique trip identifier	PK
Booking_ID	Associated booking	FK
Start_Time	Trip start time	
End_Time	Trip end time	
Distance	Distance travelled	
Fare	Final trip fare	
Trip_Status	Started / Completed / Cancelled	

PAYMENT

Attribute	Description	Key
Payment_ID	Unique payment identifier	PK
Booking_ID	Associated booking	FK
Amount	Amount paid	
Payment_Method	UPI / Card / Cash	
Payment_Date	Payment date	
Payment_Status	Successful / Pending / Failed	

RATING

Attribute	Description	Key
Rating_ID	Unique rating identifier	PK
Booking_ID	Associated booking	FK
Customer_ID	Customer giving rating	FK
Driver_ID	Driver being rated	FK
Rating	Rating from 1-5	
Review	Customer feedback	

DRIVER_LOCATION

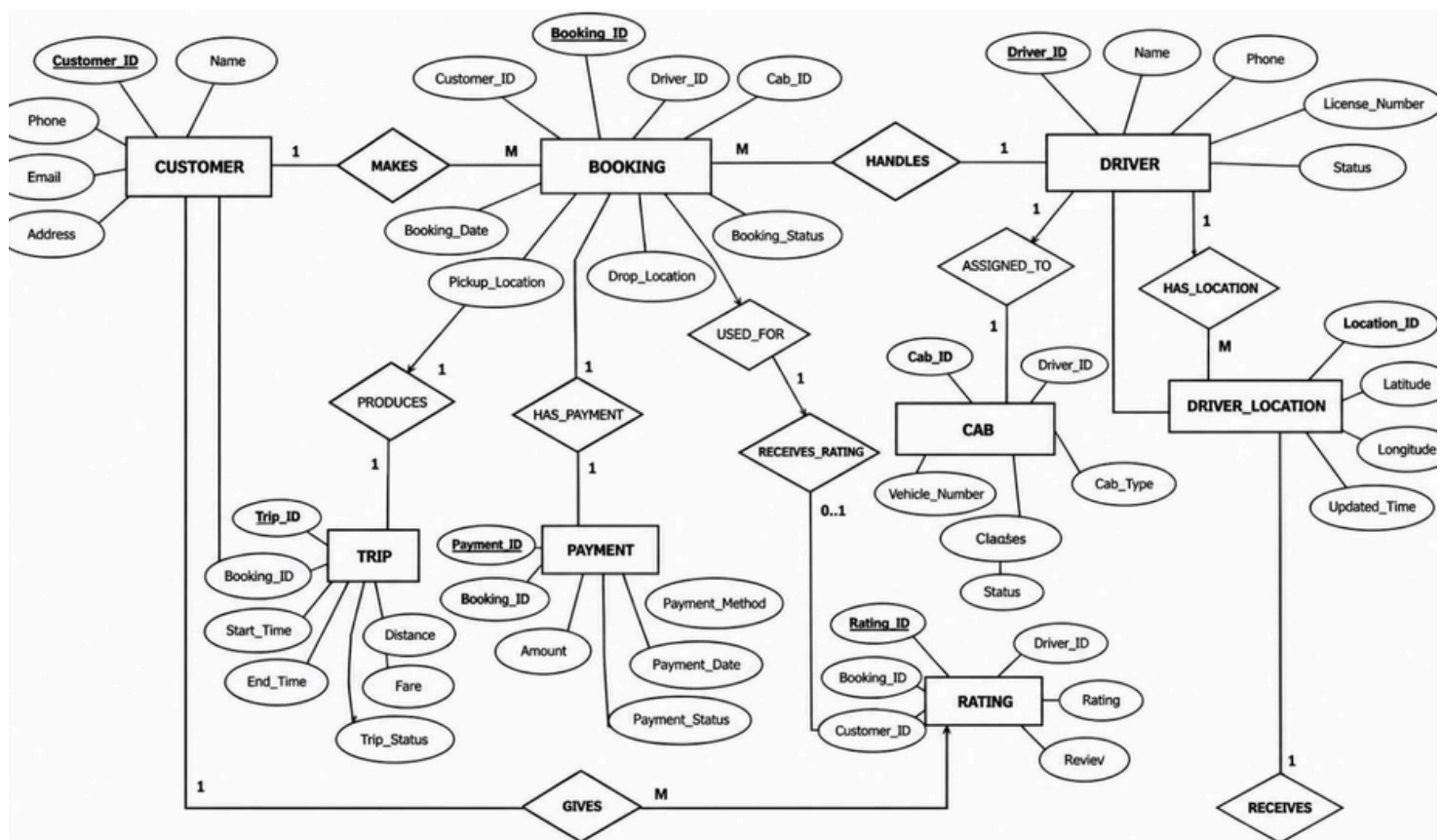
Attribute	Description	Key
Location_ID	Unique location record	PK

Attribute	Description	Key
Driver_ID	Driver being tracked	FK
Latitude	Latitude coordinate	
Longitude	Longitude coordinate	
Updated_Time	Last location update	

2. Relationships and Cardinalities

Relationship	Cardinality	Explanation
Customer → Booking	1 : M	One customer can make many bookings.
Driver → Booking	1 : M	One driver can handle many bookings over time.
Driver → Cab	1 : 1	One driver is associated with one cab in the simplified model.
Cab → Booking	1 : M	One cab can be used for many bookings over time.
Booking → Trip	1 : 1	One booking produces one trip record.
Booking → Payment	1 : 1	One booking has one payment record.
Booking → Rating	1 : 0..1	A booking may or may not receive a rating.
Driver → Driver_Location	1 : M	A driver's location can be updated many times.
Customer → Rating	1 : M	A customer can submit multiple ratings over time.
Driver → Rating	1 : M	A driver can receive multiple ratings from rides.

3. ER Diagram



4. Relational Schema

Relation	Attributes
CUSTOMER	Customer_ID (PK), Name, Phone, Email, Address
DRIVER	Driver_ID (PK), Name, Phone, License_Number, Status
CAB	Cab_ID (PK), Driver_ID (FK), Vehicle_Number, Cab_Type, Model, Status
BOOKING	Booking_ID (PK), Customer_ID (FK), Driver_ID (FK), Cab_ID (FK), Booking_Date, Pickup_Location, Drop_Location, Booking_Status
TRIP	Trip_ID (PK), Booking_ID (FK), Start_Time, End_Time, Distance, Fare, Trip_Status
PAYMENT	Payment_ID (PK), Booking_ID (FK), Amount, Payment_Method, Payment_Date, Payment_Status
RATING	Rating_ID (PK), Booking_ID (FK), Customer_ID (FK), Driver_ID (FK), Rating, Review
DRIVER_LOCATION	Location_ID (PK), Driver_ID (FK), Latitude, Longitude, Updated_Time

Foreign-key references

Foreign Key	References
CAB.Driver_ID	DRIVER(Driver_ID)
BOOKING.Customer_ID	CUSTOMER(Customer_ID)
BOOKING.Driver_ID	DRIVER(Driver_ID)
BOOKING.Cab_ID	CAB(Cab_ID)
TRIP.Booking_ID	BOOKING(Booking_ID)
PAYMENT.Booking_ID	BOOKING(Booking_ID)
RATING.Booking_ID	BOOKING(Booking_ID)
RATING.Customer_ID	CUSTOMER(Customer_ID)
RATING.Driver_ID	DRIVER(Driver_ID)
DRIVER_LOCATION.Driver_ID	DRIVER(Driver_ID)

5. Primary Key and Foreign Key Summary

Table	Primary Key	Foreign Keys
CUSTOMER	Customer_ID	—
DRIVER	Driver_ID	—
CAB	Cab_ID	Driver_ID
BOOKING	Booking_ID	Customer_ID, Driver_ID, Cab_ID
TRIP	Trip_ID	Booking_ID
PAYMENT	Payment_ID	Booking_ID
RATING	Rating_ID	Booking_ID, Customer_ID, Driver_ID

Table	Primary Key	Foreign Keys
DRIVER_LOCATION	Location_ID	Driver_ID

6. Brief Relationship Explanation

Customer — Booking (1:M)

A customer can make multiple cab bookings, while each booking belongs to one customer.

Driver — Booking (1:M)

A driver can handle multiple bookings over time, while each booking is assigned to a driver.

Driver — Cab (1:1)

Under the simplified model used in Assignment 1, one driver is associated with one cab and one cab is associated with one driver.

Cab — Booking (1:M)

A cab can be used for multiple bookings over time, while each booking is assigned to one cab.

Booking — Trip (1:1)

Each booking produces one trip record containing journey time, distance, fare and trip status.

Booking — Payment (1:1)

Each booking has a corresponding payment record containing amount, payment method, date and payment status.

Booking — Rating (1:0..1)

A booking can receive a rating, but the rating is optional because a customer may complete a ride without submitting feedback.

Driver — Driver_Location (1:M)

One driver can have many location records because the driver's location may be updated repeatedly.

Customer — Rating (1:M)

A customer can submit multiple ratings for different completed rides.

Driver — Rating (1:M)

A driver can receive multiple ratings from different rides.